USER INFORMATION
READ CAREFULLY

METRONOL®
(Metronidazole 0.5% w/v Injection)

Product Description:
Metronidazole 0.5% w/v Injection is a sterile non-pyrogenic solution. The solution is clear, almost colourless to pale yellow solution.

Composition:
100mL solution contains:-
Metronidazole BP 500mg in citrate-Phosphate buffered isotonic solution.

Electrolytes:	mmol/L	mEq/L
Sodium Ion (Na^+)	141.00	141.00
Chloride Ion (Cl^-)	135.00	135.00

Classification and Activity:
Metronidazole is a nitroimidazole anti-infective agent which has specific activity against a number of obligate anaerobic organisms and protozoa.

Pharmacodynamic:
Metronidazole is an anti-infectious drug belonging to the pharmacotherapeutic group of nitroimidazole derivatives, which have effect mainly on strict anaerobes. This effect is probably caused by interaction with DNS and different metabolites.
Metronidazole has antibacterial and antiprotozoal actions and is effective against anaerobic bacteria and against *Trichomonas vaginalis* and other protozoa including *Entamoeba histolytica* and *Giardia lamblia*.

<u>Anti-Microbial Spectrum:</u>
The MIC breakpoints separating susceptible from intermediately susceptible and intermediately susceptible from resistant organisms are as following:
Susceptible (S) = 4 mg/L and Resistant (R) > 4 mg/L
The prevalence of acquired resistance may vary geographically and with time for selected species and local information is desirable, particularly when treating severe infections. This information gives only approximate guidance on probabilities whether microorganisms will be susceptible to Metronidazole or not.

<u>Categories of microorganism susceptible to Metronidazole:</u>
Gram negative aerobes: *Helicobacter pylori*
Anaerobes: *Bacteroides fragilis, Bifidobacterium, Bilophila, Clostridium, Clostridium difficile, Clostridium perfringens, Eubacterium, Fusobacterium, Peptostreptococcus, Prevotella, Porphyromonas and Veillonella.*

<u>Categories of microorganism resistance to Metronidazole:</u>
Gram positive aerobes: *Actinomyces*
Anaerobes: *Mobiluncus, Propionibacterium acnes*

Pharmacokinetics:
Distribution: After administration of a single 500 mg dose, mean Metronidazole peak plasma concentrations of approx. 14 - 18 µg/mL are reached at the end of a 20 minute infusion. 2-hydroxy-metabolite peak plasma concentrations of approx. 3 µg/mL are obtained after a 1 g single I.V. dose. Steady state Metronidazole plasma concentrations of about 17 and 13 µg/mL are reached after administration of Metronidazole every 8 or 12 hours, respectively.
Plasma protein binding is less than 10%, and the volume of distribution 1.1 ± 0.4 L/kg.

Metabolism: Metronidazole is metabolised in the liver by hydroxylation, oxidation and glucuronidation. The major metabolites are a 2-hydroxy- and an acetic acid metabolite.

Elimination: More than 50% of the administered dose is excreted in the urine as unchanged Metronidazole (approx. 20% of the dose) and its metabolites. About 20% of the dose is excreted with faeces. Clearance is 1.3 ± 0.3 mL/min/kg, while renal clearance is about 0.15 mL/min/kg. The plasma elimination half-life of Metronidazole is approx. 8 hours, and of the 2-hydroxy-metabolite approx. 10 hours.

Special patient groups: The plasma elimination half-life of Metronidazole is not influenced by renal impairment, however this may be increased for 2-hydroxy- and an acetic acid metabolite. In the case of haemodialysis, Metronidazole is rapidly excreted and the plasma elimination half-life is decreased to approx. 2.5 hour. Peritoneal dialysis does not appear to affect the elimination of Metronidazole or its metabolites compared to patients with renal impairment.
In patients with impaired liver function, the metabolism of Metronidazole is expected to decrease, leading to an increase in the plasma elimination half-life. In patients with severe liver impairment, clearance may be decreased up to approx. 65%, resulting in an accumulation of Metronidazole in the body.

Indications:
Metronidazole 0.5% w/v Injection is indicated:
(a) For treatment of anaerobic infections in patients (adults and children) for whom oral administration is not possible.
(b) The prophylaxis of postoperative infections due to sensitive anaerobic bacteria particularly species of Bacteroides and anaerobic Streptococci, during abdominal, gynaecological gastrointestinal or colorectal surgery which carries a high risk of occurrence of this type of infection. The solution may also be used in combination with an antibiotic active against aerobic bacteria.
(c) The treatment of severe intraabdominal and gynaecological infections in which sensitive anaerobic bacteria particularly Bacteriodes and anaerobic Streptococci have been identified or are suspected to be the cause.
Note: Metronidazole is inactive against aerobic or facultative anaerobic bacteria.

Recommended Dosage:
Unless otherwise prescribed, the following dosage guidelines are recommended:
Adult: 500mg every 8 hours.
Child : 7.5mg/kg body weight every 8 hours. Neonates: 15mg/kg LD, followed by 7.5mg/kg every 12 hourly.

Dosage Individualization:
Dose is not altered with impaired renal function. Patient's with severe hepatic disease metabolize Metronidazole slowly, with resultant accumulation of Metronidazole and its metabolites in the plasma. The usual recommended dose may eventually have to be reduced.

Route of Administration: Intravenous (IV)
Metronidazole 0.5% w/v Injection should be infused intravenously at an approximate rate of 5 mL/minute (or one bottle infused over 20 to 60 minutes). Oral medication should be substituted as soon as feasible.

Contraindications:
Metronidazole is contraindicated:
(a) In patients who are hypersensitive to Metronidazole or other imidazole derivatives or any of the excipients.
(b) In the first trimester of pregnancy.
(c) In patients with end stage liver damage, haematopoietic disorders and uncontrolled diseases of the central or peripheral nervous system.

Warning and Precautions:
<u>Liver Disease:</u> Caution is needed in patients with severe hepatic impairment. The dose of Metronidazole should be reduced as necessary. Metronidazole is mainly metabolised by hepatic oxidation. Substantial impairment of Metronidazole clearance may occur in the presence of advanced hepatic insufficiency. The risk/benefit ratio of using Metronidazole to treat trichomoniasis in such patients should be carefully considered. Plasma levels of Metronidazole should be closely monitored.

Caution is needed in patients with hepatic encephalopathy. Patients with severe hepatic encephalopathy metabolize Metronidazole slowly, with resultant accumulation of Metronidazole. This may cause exacerbation of CNS adverse effects. The dose of Metronidazole should be reduced as necessary.

Cases of severe hepatotoxicity/acute hepatic failure, including cases with a fatal outcome with very rapid onset after treatment initiation in patients with Cockayne syndrome have been reported with products containing Metronidazole for systemic use. In this population, Metronidazole should therefore be used after careful benefit-risk assessment and only if no alternative treatment is available. Liver function tests must be performed just prior to the start of therapy, throughout and after end of treatment until liver function is within normal ranges, or until the baseline values are reached. If the liver function tests become markedly elevated during treatment, the drug should be discontinued.
Patients with Cockayne syndrome should be advised to immediately report any symptoms of potential liver injury to their physician and stop taking Metronidazole.

<u>Active Central Nervous System Disease:</u> Metronidazole should be used with caution in patients with active disease of the Peripheral and Central Nervous System. Severe neurological disturbances (including seizures and peripheral and optic neuropathies) have been reported in patients treated with Metronidazole. Stop Metronidazole treatment if any abnormal neurologic symptoms occur such as ataxia, dizziness, confusion or any other CNS adverse reaction. The risk of aggravation of the neurological state should be considered in patients with fixed or progressive paraesthesia, epilepsy and active disease of the central nervous system except for brain abscess.
Encephalopathy has been reported in association with cerebellar toxicity characterized by ataxia, dizziness, dysarthria, and accompanied by CNS lesions seen on magnetic resonance imaging (MRI). CNS symptoms and CNS lesions, are generally reversible within days to weeks upon discontinuation of Metronidazole.
Aseptic meningitis can occur with Metronidazole. Symptoms can start within hours of dose administration and generally resolve after Metronidazole therapy is discontinued.

<u>Blood Dyscrasias:</u> Metronidazole should be used with caution in patients with evidence or history of blood dyscrasia as agranulocytosis, leukopenia and neutropenia have been observed following Metronidazole administration.

<u>Renal Disease:</u> Metronidazole is removed during haemodialysis and should be administered after the procedure is finished.
Patients with renal impairment, including patients receiving peritoneal dialysis, should be monitored for signs of toxicity due to the potential accumulation of toxic Metronidazole metabolites.

<u>Sodium restricted patients:</u> This medicinal product contains sodium. To be taken into consideration by patients on a controlled sodium diet.

<u>Alcohol:</u> Patients should be advised to discontinue consumption of alcoholic beverages or alcohol-containing products before, during, and up to 72hours after taking metronidazole because of a disulfram-like effect (abdominal cramps, nausea, headaches, flushing, vomiting and tachycardia).

<u>Intensive or prolonged Metronidazole therapy:</u> As a rule, the usual duration of therapy with IV Metronidazole or other imidazole derivatives is usually less than 10 days. This period may only be exceeded in individual cases after a very strict benefit-risk assessment. Only in the rarest possible case should the treatment be repeated. Limiting the duration of treatment is necessary because damage to human germ cells cannot be excluded.

Intensive or prolonged Metronidazole therapy should be conducted only under conditions of close surveillance for clinical and biological effects and under specialist direction. If prolonged therapy is required, the physician should bear in mind the possibility of peripheral neuropathy or leucopenia. Both effects are usually reversible. In case of prolonged treatment, occurrence of undesirable effects such as paraesthesia, ataxia, dizziness and convulsive crises should be checked. High dose regimes have been associated with transient epileptiform seizures.
Monitoring: Due to increased risk for adverse reaction, regular clinical and laboratory monitoring (including blood count) are advised in cases of high-dose, prolonged or repeated treatment, in case of antecedents of blood dyscrasia, in case of severe infection and in severe hepatic insufficiency.
General: Patients should be warned that Metronidazole may darken urine (due to Metronidazole metabolite).
Special Precaution for Handling: THE SOLUTION SHOULD BE ADMINISTERED WITH STERILE EQUIPMENT USING AN ASEPTIC TECHNIQUE.

Interactions with Other Medicaments
Concomitant therapy requiring special precautions:
(a) **Oral anticoagulants (warfarin):** Increase of the effects of oral anticoagulants and the risk of haemorrhage (decrease in its liver catabolism). Prothrombin time should be monitored more frequently. The dose of oral anticoagulants should be adjusted during the treatment with Metronidazole and 8 days after withdrawal. Hence, close monitoring of prothrombin time and International Normalized Ratio (INR) in this group of patient is highly recommended.
A large number of patients have been reported showing an increase in oral anticoagulant activity whilst receiving concomitant antibiotic therapy. The infectious and inflammatory status of the patient, together with their age and general well-being are all risk factors in this context. However, in these circumstances it is not clear as to the part played by the disease itself or its treatment in the occurrence of prothrombin time disorders. Some classes of antibiotics are more likely to result in this interaction, notably fluoroquinolones, macrolides, cyclines, cotrimoxazole and some cephalosporins.
(b) **Vecuronium (non depolarising curaremimetic):** Metronidazole can potentialise the effects of vecuronium.

Combinations to be considered:
(a) **5 Fluoro-uracile:** Increase in the toxicity of 5 fluoro-uracile due to a decrease of its clearance.
(b) **Lithium:** Lithium retention accompanied by evidence of possible renal damage has been reported in patients treated simultaneously with lithium and Metronidazole. Lithium treatment should be tapered or withdrawn before administering Metronidazole. Plasma concentrations of lithium, creatinine and electrolytes should be monitored in patients under treatment with lithium while they receive Metronidazole.
(c) **Barbiturates:** Phenobarbital might induce the metabolism of Metronidazole, which could lead to decreased efficacy of Metronidazole.
(d) **Cholestyramine:** May delay or reduce the absorption of Metronidazole.
(e) **Phenytoin:** Concomitant administration of phenytoin and Metronidazole may affect the metabolism of Metronidazole.
(f) **Cimetidine:** Inhibits the metabolism of Metronidazole.
(g) **Cyclosporine:** Case reports indicate that concomitant treatment with Metronidazole and Cyclosporine might lead to increased serum levels of cyclosporine. Cyclosporine concentrations and creatinine levels should be monitored.
(h) **Busulfan:** Plasma concentrations of busulfan may increase during concomitant treatment with Metronidazole, which can result in serious busulfan toxicity.

Pregnancy and Lactation:
Pregnancy: Clinical data on a large number of exposed pregnancies and animal data did not show a teratogenic or foetotoxic effect. However unrestricted administration of nitroimidazolene to the mother may be associated with a carcinogenic or mutagenic risk for the unborn or newborn child. Therefore Metronidazole should not be given during pregnancy unless clearly necessary. Metronidazole is contraindicated in the first trimester of pregnancy.
Lactation: Metronidazole is excreted in breast milk. During lactation either breast-feeding or Metronidazole should be discontinued.

Adverse Effects / Undesirable Effect
Common undesirable effects: Gastrointestinal tract: diffuse symptoms of intolerance (like nausea, vomiting), metallic taste, stomatitis and glossitis and dry mouth; myalgia.
Uncommon undesirable effects: Leucopenia, headaches and weakness.
Rare undesirable effects:
(a) **General:** fever, skin rashes, urticaria, erythema multiforme anaphylactic shock, Quincke oedema, pustolosis, Stevens Johnson Syndrome and Toxic Epidermal Necrolysis.
(b) **Neurology:** drowsiness, dizziness, ataxia, peripheral neuropathy or transient epileptiform seizures, hallucinations, Encephalopathy, optic neuropathy and aseptic meningitis.
(c) **Blood:** agranulocytosis, neutropenia, thrombocytopenia, pancytopenia. Blood dyscrasia is generally reversible but fatal cases have been reported.
(d) **Liver:** Abnormal function tests, cholestatic hepatitis jaundice, pancreatitis; rare and reversible cases of pancreatitis are reported.
(e) **Gastrointestinal:** Mucositis, epigastralgia, nausea, vomiting, diarrhoea, anorexia.
(f) **Urine:** darkening of urine.
(g) **Eyes:** diplopia, myopia.
(h) Herxheimer reaction
(i) Changes in the blood picture as well as peripheral neuropathy observed after prolonged treatment or high dosages generally abate after treatment withdrawal.

Overdose and Treatment:
Symptoms: In cases of overdose in adults, the clinical symptoms are usually limited to nausea, vomiting, ataxia and slight disorientation. In a preterm newborn, no clinical or biological sign of toxicity developed.
Treatment: There is no specific treatment for Metronidazole overdose, Metronidazole infusion should be discontinued. Patients should be treated symptomatically.

Effects on Ability to Drive and Use Machine:
No studies have been performed following intravenous treatment with Metronidazole on the ability to drive and use machines. Some adverse reactions to Metronidazole such as seizure, dizziness, optic neuropathy, may impair the ability to drive or operate machines.
Therefore, it is recommended that patients should not drive or use machines.

Incompatibilities:
In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal product.

Storage Conditions:
Do not store above 30°C. Protect from light.

Shelf Life:
The product has a shelf life of 3 years from manufacturing date in the proper storage condition in LDPE Bottle.

Dosage Forms and Packaging Available:
100mL X 20 LDPE plastic bottle per Carton.

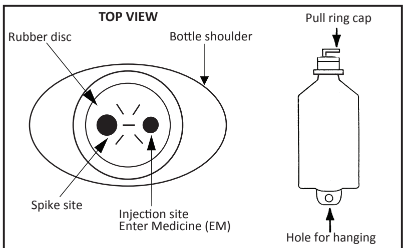
**** Keep out of reach of children/jauhkan daripada kanak-kanak***

Manufacturer / Product Registration Holder:
AIN MEDICARE SDN.BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

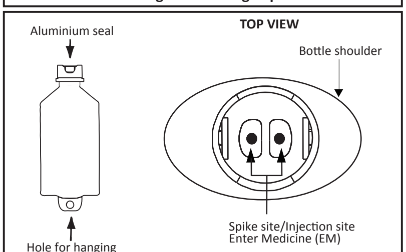
HANDLING INSTRUCTION

AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE
The bottle is made from injectable grade polyethylene.

PARTS OF THE PLASTIC BOTTLE

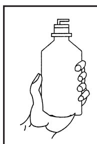


Design 1: Pull Ring Cap



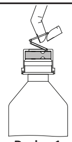
Design 2: Twin Port Cap (Aluminum Seal)

Top view after pull ring cap is removed. Please refer whichever applicable design.
INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG


Design 1

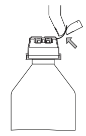
1. Before use check for any solid particle, growth, turbidity or leakage. Discard the product if particle, growth, turbidity or leakage is found.

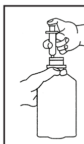

Design 2


Design 1

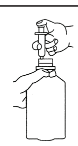
2. Design 1: Pull the cap's ring until completely removed.

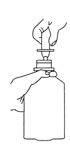
   Design 2: Pull the aluminum seal until completely removed by centre line.


Design 2


Design 1

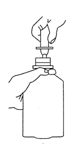
3. Additives may be injected through injection site (Refer to Design 1 or 2 whichever applicable) on the rubber. Mix the solution thoroughly.


Design 2


Design 1

4. Hold bottle down firmly and insert a spike of administration set into the spike site for Design 1 OR for Design 2, pull the other aluminium seal until completely removed before inserting a spike of administration set.

(Spike site is meant for spiking of the administration set and is only for single use)


Design 2

Date of Revision : 01.07.2019